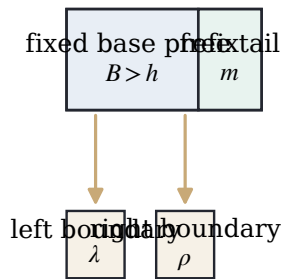
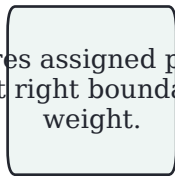


a

Tail-panel encoding



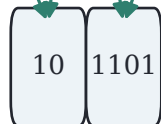
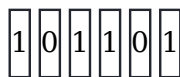
Boundary debt stores assigned pair vector, fixed left boundary bits, current right boundary bits, and remaining weight.



b

Dyadic cellular stack

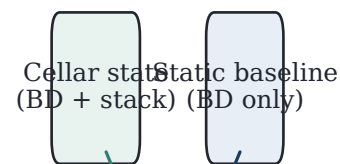
incoming suffix bits



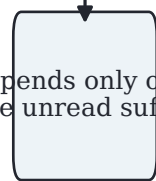
Read one bit, push a length-1 cellular, then merge equal-size trailing blocks.

c

Matched comparator and exact oracle



Same tokenization, same exact tail-completion oracle, same canonical accounting, same budget.



Exact completion depends only on boundary debt plus the unread suffix.

$$c_{axy}(k) = \alpha_x(k) + \beta_k^{\text{cross}}(\rho_x, y) + \beta_k^{\text{wrap}}(\lambda, y) + \beta_k^{\text{int}}(y)$$